

LESSON 2-10 • SESSION 2 • PREREQUISITE PRACTICE A • SKILL 1 OF 2

Describe the Shape of Data

Builds on Lesson 2-2 Session 2 • 6.SP.A.2

TODAY'S LESSON

In Problem 2, tall bars on the left with a long tail to the right mean the data is skewed right.

YOU NEED TO

Describe a data set as symmetric, skewed right or skewed left. **You will use it in Practice Form A, problems 2 and 8.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1 Find** where most values pile up.
- 2 Look** for a long tail of a few values on one side.
- 3 Name** it: tail right, skewed right; tail left, skewed left; no tail, symmetric.

WATCH ME*the steps in action***Describe 2, 3, 3, 4, 4, 4, 15.**

- Step 1** Most values pile up at 2 to 4.
Step 2 One value, 15, makes a long tail to the right.
Step 3 Tail right: skewed right.

Answer skewed right**TRY IT WITH HELP***fill in each step*

1. Describe 1, 9, 10, 10, 11, 11, 11.
Step 1 Most values pile up at _____ to 11.
Step 2 The long tail is on the _____.
Step 3 Shape: _____
2. Describe 4, 5, 5, 6, 7, 7, 8.
Step 1 The values pile up in the _____.
Step 2 Is there a long tail? _____
Step 3 Shape: _____

YOUR TURN*use the steps on your own*

3. Write symmetric, skewed right or skewed left.
 - a. 31, 33, 35, 37, 39 _____
 - b. 12, 13, 15, 17, 19, 21, 400 _____
 - c. 2, 40, 42, 44, 45 _____
 - d. 44, 46, 48, 50, 52, 54, 56 _____

Find the IQR and Range

Builds on Lesson 2-5 Session 1 • 6.SP.B.5c

TODAY'S LESSON

In Problem 4 the ticket sales are skewed, so you report the median and the IQR.

YOU NEED TO

Find the range and the interquartile range (IQR) of a data set. **You will use it in Practice Form A, problems 4 and 8.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1 Order** the values. Range = greatest – least.
- 2 Find** Q1 and Q3, the medians of the lower and upper halves.
- 3 Subtract:** IQR = Q3 – Q1.

WATCH ME*the steps in action*

Find the range and IQR of 12, 13, 15, 17, 19, 21, 400.

Step 1 Range = $400 - 12 = 388$.

Step 2 Lower half 12, 13, 15: Q1 = 13. Upper half 19, 21, 400: Q3 = 21.

Step 3 IQR = $21 - 13 = 8$.

Answer range **388**, IQR **8**

TRY IT WITH HELP*fill in each step*

4. Find the range and IQR of 2, 4, 6, 8, 10, 12, 14.

Step 1 Range = $14 - 2 =$ _____

Step 2 Q1 = _____. Q3 = _____

Step 3 IQR = $12 - 4 =$ _____

5. Find the range and IQR of 5, 10, 15, 30, 45.

Step 1 Range = $45 - 5 =$ _____

Step 2 Q1 = _____. Q3 = _____

Step 3 IQR = _____

YOUR TURN*use the steps on your own*

6. Find the IQR.

a. 1, 3, 5, 7, 9, 11, 13 _____

b. 10, 20, 30, 40, 50, 60, 70 _____

c. 2, 4, 6, 8, 10 _____

d. 44, 46, 48, 50, 52, 54, 56 _____

READY CHECK*a problem like the one in today's lesson*

7. For 15, 18, 20, 24, 30, 35, 41, Q1 = 18 and Q3 = 35. What is the IQR?

A 17

B 26

C 24

D 53

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____